

# Hany Hamed Aly

Interested in Robot Learning, Multi-Robot Systems & Sim2Real

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## EDUCATION

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### Innopolis University

Innopolis, Russia

*Bachelor of Computer Science (Robotics track) with **Full scholarship***

*Aug 2018 – June 2022*

- Outstanding Contribution to Science at Innopolis University during 2020/2021 ([Certificate](#))
- Outstanding Achievements at Innopolis University for Fall 2020 ([Diploma](#))

**CGPA:** 4.65/5 ([Original Transcript](#))

## RESEARCH EXPERIENCE

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### Bachelor Thesis | *In progress*

May 2021 – June 2022

- Topic: **Learning behavioural strategies for a multi-robot system in a predator-prey environment using Reinforcement Learning**
- Keywords: reinforcement learning, predator-prey, self-play, multi-agent
- Progress: [link](#)
- Supervised by: Prof. [Stefano Nolfi](#) & Prof. [Alexandr Klimchik](#)

### Laboratory Assistant

June 2021 – Present

*Unmanned Technology Laboratory, Innopolis University*

*Innopolis, Russia*

- Integrating and developing ROS packages of different modules for drones (Gimbal control, RTK GPS, ...etc)
- Integrated livox lidar with a drone for outdoor mapping
- Developed a handheld lidar for indoor/outdoor mapping for an industrial partner ([results](#))

### Undergraduate Research Assistant

Aug 2019 – April 2021

*Center for Technologies in Robotics and Mechatronics Components, Innopolis University*

*Innopolis, Russia*

- Implemented a gym environment for a Tensegrity hopper
- Developed experiments using ARS RL algorithm to learn a stabilizing control policy for a tensegrity hopper
- Designed and implemented a contactless differentiable physics simulator for tensegrity robots using Taichi
- Performed research on sim2real transfer for a three-prism tensegrity robot

## PUBLICATIONS & IN PROGRESS MANUSCRIPTS

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- G. Kulathunga, H. Hamed, D. Devitt, and A. Klimchik, "Optimization-based trajectory tracking approach for multi-rotor aerial vehicles in unknown environments," *IEEE Robotics and Automation Letters*, 2022 ([IEEE paper](#), [arXiv](#))
- V. Kurenkov, H. Hamed, and S. Savin, "Learning stabilizing control policies for a tensegrity hopper with augmented random search," in *2020 International Conference on Industrial Engineering, Applications and Manufacturing (ICIEAM)*, pp. 1–5, IEEE, 2020 ([code](#), [paper](#))
- L. Vorochaeva, S. Savin, H. Hamed, and A. M. Leon, "Analysis of algorithms for controlling the length of crawling robot modules," in *2020 4th Scientific School on Dynamics of Complex Networks and their Application in Intellectual Robotics (DCNAIR)*, pp. 257–260, IEEE, 2020 ([paper](#))
- L. Vorochaeva, S. Savin, and H. Hamed, "Lateral gait analysis of a crawling robot by means of controlling the lengths of links and friction in the supports," in *2020 International Conference Nonlinearity, Information and Robotics (NIR)*, pp. 1–6, IEEE, 2020 ([paper](#))

- H. Hamed, V. Kurenkov and S. Savin, “Differentiable Tensegrity Simulator with Sim2Real Experiments” [Manuscripts in preparation]
- H. Hamed, S. Savin and L. Vorochaeva “A Brief Survey on Locomotion Control Methods For Crawling Robots” [Manuscripts to be submitted in Nonlinearity, Information and Robotics (NIR) 2022]

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## SELECTED AWARDS & CERTIFICATES & SUMMER SCHOOLS

- Online Asian Machine Learning School (OAMLS 2021, [certificate](#))** November 2021
- This school covered multiple topics in Machine learning: computer vision, deep learning, NLP, representation learning, transfer learning, reinforcement learning, causality and meta learning.
- European Agent Systems Summer School (EASSS 2021)** July 2021
- The summer school explored different topics related to multi-agent systems around cooperative systems, reinforcement learning, game theory, modeling and simulation.
- 3rd place DOTS competition from Bristol Robotics Lab and Toshiba ([Certificate](#))** June 2021
- The competition was about developing an algorithm for multi robots to collect food items and return it to the storage in a gazebo simulated environment.
- Winter School on Machine Learning in Robotics ([Certificate of Participation](#))** Dec 2020
- This school explored the intersection of machine learning in robotics in areas of reinforcement learning, and computer vision

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## SKILLS & TOOLS

- ROS, Python, C/C++, PyTorch, Docker, git, Linux

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## WORKING EXPERIENCE

- Technical Intern** July 2019 – Aug 2019 - [[Doc](#)]  
*Copter Express* *Moscow, Russia*
- Had a training about main components and construction of drones
  - Had a training on programming COEX's drones (ROS, px4 and indoor navigation using Aruco markers)
  - Developed a human pose estimation with a webcam using Tensorflow.js to control a drone
  - Integrated RTAP-Map SLAM algorithm with COEX's drone using Intel Real-Sense T265 and D435 cameras

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## PROJECTS

- Drone Detection (Computer Vision project course) | [Link](#) | *PyTorch*** September 2021
- Collected and labeled a diverse dataset
  - Implemented training scripts for Faster R-CNN pretrained models from detectron2
- Human Pose Estimation Drone Control | [Link](#) | *Python, TF.js, ROS*** July 2019
- Explored and tested different human pose estimation modules
  - Integrated a drone controller with human pose estimation module built with TensorFlow.js

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## EXTRACURRICULAR ACTIVITIES

- Judge at WRO International Finals [Advanced Robotics Challenge - ARC] | [Certificate](#)** 2019
- Developed the game rules for the international competition
  - Judged the Russian qualifications as the Head judge and the International competition as a judge
  - Collaborated in the development of a playground generator for the competition using C++ and Javascript
- Participant in RoboCup Junior Egypt Open Weight Soccer Category (2nd place)** 2018
- Developed ball detection using OpenCV and the robot's game controller algorithm
  - Developed PCB boards for the robot